DEPARTMENT OF HEALTH AND HUMAN SERVICES

Office of the Secretary

RFI to Harness Artificial Intelligence
To Deflate Health Care Costs and Make
America Healthy Again (Dec. 19, 2026)

RFI: Accelerating the Adoption and Use of Artificial
Intelligence as Part of Clinical Care (Dec. 23, 2026)

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The Dimensions of Artificial Intelligence in the Healthcare Industry

The Department of Health and Human Services ("HHS") should be commended for seeking to leverage technology to lower healthcare costs and improve care. Since being diagnosed with Parkinson's disease 24 years ago, I have watched healthcare R&D – across multiple disease fronts – proceed at a glacial pace. Conversely, costs (and related subsidies) have skyrocketed.

Fortuitously, artificial intelligence offers the potential to fundamentally change these trends. HHS should view AI's short-term effect as improving the odds of success and shortening feedback cycles. However, to achieve significant breakthroughs as AI systems develop, HHS (via the FDA) will need to completely revamp its approach to drug discovery and development. Rather than being an overzealous gatekeeper at every step of the way, its role should shift to being an auditor with specifically delimited responsibilities.

Dimension #1. Incorporation of AI into Drug Discovery

The biggest benefit to the healthcare industry's performance from AI is achievable from drug discovery. The federal government expends $44 billion per year in funding basic healthcare research. However, a small fraction of basic research initiatives pass through the "Valley of Death" and make it to a clinical trial. The average length of this phase is approximately 15 years per endeavor, where FDA involvement impacts the pace of progress. Even after overcoming this initial barrier, less than 10% of such promising remedies receive final FDA approval.

Accounting for the costs of failures, the average FDA drug approval costs society almost $3 billion and takes decades to reach the market from its inception in the lab. This system is a prescription for failure, and the results of such a scheme appear every day in the real world in the form of high costs and poor performance.

In contrast, AI identifies potential treatments much faster than traditional methods. By enabling the efficient exploration of biological variability, AI can dramatically increase the number of experiments by studying up to a trillion interactions between variables. In other words, AI can process vast amounts of biological data, uncover hidden causal relationships, and generate new actionable insights.

Significantly, more than 50% of the nation's healthcare costs are attributable to patients with multiple chronic illnesses and this figure exceeds 75% when complex diseases are included. AI is particularly promising for complex, multifactorial conditions – such as neurodegenerative diseases, autism spectrum disorders, and multiple chronic illnesses – where conventional reductionist approaches have failed.

Additionally, there are over 30 million people in the U.S. affected by rare diseases, with more than 7,000 different types in existence.

HHS should encourage AI initiatives to address these conditions because they hold the potential to improve the health of the nation's most disadvantaged citizens and significantly cut the costs of providing care. Let the scientists study science, not the FDA requirements. By being forced to have one eye on the finish line, scientists tend to craft their work to satisfy the FDA, rather than the rudiments of a flawed biological process.

In the short-run, HHS should direct its grants toward AI-generated basic research, with a particular emphasis on the hard-to-solve illnesses. At the same time, the FDA should be putting into place a new (de)regulatory system for AI-initiated programs to enable breakthrough treatments in a compressed timetable.

Dimension #2. Incorporation of AI into the Drug Development Process

Simply relying on AI for drug discovery, while subjecting its advances to the current approval process would undermine the use of the technology. The current clinical procedures employed by the FDA entail an average of 12 years and $2 billion for each drug approval. Even scheduling a meeting can take six months for a "fast track" application. Enhancements to the existing processes alone would not capture the true potential of AI.

As a start, improvements from AI can already be had in fulfilling the exhaustive regulatory documentation requirements, which today add up to as much as 30% of the cost of compliance.

In the short-run, AI can improve drug development by:

- Automating and validating regulatory documentation

- Enhancing trial design and participant stratification
- Monitoring safety and efficacy in near real-time
- Reducing administrative and compliance costs

For example, in the U.K., the Medicines and Healthcare Products Regulatory Agency reported that clinical trial approval times were twice as fast with AI and associated reforms.

To achieve much greater long-term gains, HHS should not be involved in pre-clinical AI research and should collapse all clinical work utilizing AI into one elongated trial rather than discrete Phase I, II and III trials, given that AI can be used to continually update and validate documentation. This change would not require statutory change or agency rulemaking because clinical trial design is not codified in the FDA's rules.

As participants are added to a trial, safety results can be examined and reported in real time. Once the trial surpasses a certain number such as 1000 participants with proven efficacy and meeting the specified safety protocols, it would be approved for roll-out. The role of the government in such an approach would be as auditor to validate the output of the trial. This function would include experimental validation, mechanistic understanding, and ethical oversight. Real-world data also could more readily collected after launch thereby offering another safety feature.

With these changes, FDA personnel would shift from episodic gatekeepers to continuous auditors, which would require a fundamental change in organizational culture. While safety concerns would remain important, the responsibility and accountability would be more equally shared with the applicants and trial participants. Additionally, the prolonged suffering of existing patients would be factored into the public welfare analysis in reviewing preliminary safety results. For example, the 33 million people with one or more of the 7000 rare diseases now have a near 0% chance of relief under current regulatory procedures, and their plight must be considered while the contours of biology are examined.

A specific example of this imbalance relates to the AI-generated treatment of Familial Adenomatous Polyposis ("FAP"). The new treatment produced a large, sustained reduction in polyps with no significant safety concerns.

The company hopes to meet with the FDA in 1H2026. In the meanwhile, the current treatment for FAP is an expensive, intrusive, and a lifelong endeavor. Early detection strategies cost $10k+ and late detection costs $37k+. The cost to treating metastatic colorectal cancer (for which FAP predisposes) can be extremely high, up to $300,000. The constant exams and excisions are the norm for these patients. Yet the FDA is not inclined to ask these patients whether they would like to try the pill or whether continued excisions would be preferable. In this case, the answer from most patients would be obvious – if given the choice.

Dimension #3. Enhance Data Collection to Empower AI

AI harbors the proverbial statistical stigma of "garbage in, garbage out." Accordingly, comprehensive, and accurate, data is essential to its success. Yet this another area where the healthcare industry has failed.

The industry has evolved with each provider, or family of providers, encouraging their patients to sign up for a customer portal. The providers typically treat the information on those portals as their own for purposes of research. However, the providers do not own the data. Each patient owns his or her data.

There have been few instances where comprehensive data has been collected. For example, in 2010, The Michael J. Fox Foundation launched the Parkinson's Progression Markers Initiative (PPMI) to find the biological markers of Parkinson's onset and its progression. That study led to the impressive finding of a tool that can detect pathology not only of people diagnosed with Parkinson's, but also in individuals that are at a high risk of developing it. However, that study was geared towards Parkinson's disease and has only a few thousand participants.

To broaden the scope and applicability of healthcare data, HHS should establish national standards for patient-facing data collection that:

- Use interoperable formats
- Capture both diagnostic outcomes and relevant explanatory variables
- Preserve patient ownership and informed consent
- Enable longitudinal tracking while protecting privacy and security

As the above information from participants is collected over time, researchers would use AI to identify combinations of variables that point towards promising results.

Once this format is established, HHS should establish a goal of enrolling 100,000 participants within two years.

Dimension #4. Use of AI to Establish Standards of Care and Price Ceilings

There are no national standards of care for diseases or other health maladies in the United States. Patients oftentimes do not understand the nature of their infliction, the options to treat it, or the costs of the various options to remedy it.

On a parallel track, HHS might fund basic research targeted to a particular ailment, the FDA might (or might not) approve it, Medicare might (or might not) cover it, and some insurance companies may cover the treatment and some may not.

Moreover, the costs of various treatments may vary greatly from facility to facility – unbeknownst to the patient.

Layered on top of this market dysfunction, healthcare practitioners have the desire (and the economic incentive) to provide the best (and likely the most expensive) possible service to their patients.

As an economist would conclude, the supply curves and demand curves in the healthcare industry do not lead to a pareto optimal solution. That is, there is a market failure, primarily relating to a lack of actionable information.

In the short-run, AI can help address these failures by aggregating and analyzing how care is delivered across the country and identifying patterns associated with better outcomes and lower costs. These insights could be used to inform evidence-based minimum standards of care and improve transparencies around pricing and performance.

Over the longer term, the outputs of these systems could be used to establish a minimum standard of care for all (or most) ailments. These standards would be mandatorily covered by insurance. Concurrently, the outputs for these standards of care could be supplemented by regional price ceilings for the various practices based on a comprehensive industry analysis.

As experience is gained from these informational AI systems, a future version could be programmed to automatically calculate the prescribed minimum standards of care and the price ceilings to mimic the functioning of demand and supply curves. An algorithm could be constructed using a specified level of subsidy provided by the federal government as the equilibrium. As the federal subsidy exceeds certain pre-set limits, AI would be used to address the disequilibrium by providing to law makers various options that would lower the price ceiling for certain conditions and/or lower the minimum standard of care.

In scenarios where the stipulated federal subsidy was exceeded, some classes of patients would be denied receiving payment for the best available treatment (unless they had supplemental insurance) and/or some healthcare providers would suffer a diminution of profits.

Such an approach would require Congressional approval, but such tradeoffs are occurring now – without factual inputs or informed choices. In this dimension, AI could be used to address the industry's massive information failure and tackle the ever-increasing subsidies.

Dimension #5. Incorporation of AI into HHS's Internal Processes

AI can also improve the efficiency and effectiveness of HHS's internal operations. While the potential percentage gains would be smaller than that for the drug discovery and development dimensions, even modest improvements can yield meaningful savings given the scale of federal healthcare spending.

Early applications of AI should focus on areas with clear return on investment and limited risk, such as

- Fraud, waste, and abuse detection
- Claims auditing and compliance monitoring
- Internal workflow automation

HHS should deploy these tools incrementally, evaluate their performance, and use lessons learned to inform broader adoption across its various divisions.

Dimension X – "Accelerating" the Adoption of "AI" as Part of Clinical Care

HHS also issued a companion RFI on Dec. 23 seeking to accelerate the adoption of AI in clinical care. HHS should proceed very carefully on the issues raised in this RFI. For example, the AI software embedded in my browser defines clinical care as:

> Clinical care refers to the direct, hands-on medical treatment, diagnosis, and management of patients by trained healthcare professionals.

In this regard, HHS should not involve itself in anything beyond establishing standards of care for several reasons.

First, in an expanded capacity, HHS is likely to slow down the adoption of AI. For example, a few years ago, HHS recognized the need to reorganize the Center for Evaluation and Drug Research. After a multiple-year effort, it described its reorganization as:

> "The office is now comprised of six cross-functional support offices and eight clinical offices. The six cross-functional support offices oversee 12 divisions and several staff groups. The eight clinical offices oversee OND's 27 clinical divisions and six pharm/tox review divisions".

This is not the sort of thinking or organizational design to accelerate the adoption of new technology in clinical care or elsewhere.

Second, we are in the first inning of AI system development. HHS should not be in the business of accelerating or selecting any one AI system or application. Rather, it should set a level playing field and let the market decide the best technologies.

Third, HHS's legal authority does not extend to anything and everything related to healthcare.

Rather, HHS derives its authority from a series of laws. For example, the Public Health Service Act provides the authority for HHS to lead responses to public health emergencies, declare emergencies, manage the Strategic National Stockpile, and control diseases.

HIPAA (Health Insurance Portability and Accountability Act) authorizes HHS to set standards for electronic health information exchange, privacy, and security.

Other statutes govern specific areas such as giving authority to agencies like SAMHSA for substance abuse, OHRP for human research, and the OIG for oversight.

Proactively selecting or pushing a particular technology for patient care would stretch HSS's legal authority and could very well retard advancements as result of extended litigation. There are multiple AI systems "in the works." HHS should not be "in the business" of picking winners and losers through a program of acceleration.

Conclusion

AI offers the opportunity for significant improvements in healthcare outcomes and efficiencies—but only if it is integrated into a regulatory and governance framework designed for its capabilities. Shoehorning AI into existing structures will blunt its impact and increase the risk of implementation.

Each dimension described above requires a separate dedicated, multidisciplinary team reporting to the Office of the Deputy Secretary. After the strategic direction for each dimension is established, these teams should be tasked with:

- Developing detailed implementation plans, including budgetary requirements
- Identifying any statutory or regulatory barriers
- Establishing timelines, milestones, and evaluation criteria
- Addressing ethical and equity considerations

Drug discovery and drug development represent the highest-impact dimensions for AI implementation. HHS should make use of *external expertise* in fashioning the details of an appropriate regulatory framework for these dimensions.

The detailed plans for implementing AI should be approved and finalized before the end of 2026. As described herein, HHS should take a proactive, forward-looking role in harnessing AI technology to deflate healthcare costs and make America healthy again.

Sincerely,

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